Table 11.2 When and where reasoning is employed and used in various applications

Application	Implemented	Environment	Purpose	Timing	Configuration
Personal assistant	Within the LLM	Personal device	Breaking down work into steps	During the response	As part of the prompt or LLM
Customer service bot	Not typical; usually just informational	Limited tool use and need for composite tool use			
Autonomous agent	As part of the agent prompt and within the LLM	Server or service	Facilitate complex tool use and task planning	As part of LLM, external reasoning not well suited	Within the agent or LLM
Collaborative workflows	As part of the LLM	Shared canvas or coding	Assists in breaking work down	During the response	Within the LLM
Game AI	As part of the LLM	Server or application	Essential for undertaking complex actions	Before or during the response	Within the LLM
Research	Anywhere	Server	Understand how to solve complex problems and engage in complex task workflows	Before, during, and after response generation	Anywhere

Table 11.2 shows several varied application scenarios in which we may find an assistant or agent deployed to assist in some capacity. To provide further information and guidance, this list provides more details about how reasoning may be employed in each application:

- *Personal assistant*—Depending on the application, the amount of reasoning an agent employs may be limited. Reasoning is a process that requires the LLM to think through a problem, and this often requires longer response times depending on the complexity of the problem and the extent of the prompt. In many situations, responses intended to be